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Supplemental Material

The Association between Long-Term Air Pollution and Urinary Catecholamines: Evidence from the Multi-Ethnic Study of Atherosclerosis

Anjum Hajat, Ana V. Diez Roux, Cecilia Castro-Diehl, Kristen Cosselman, Sherita Hill Golden, Marnie F. Hazlehurst, Adam Szpiro, Sverre Vedal, and Joel D. Kaufman

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Figure S1. Directed Acyclic Graph (DAG) of association between air pollution and catecholamines.

Table S1: Percent differences in stress hormones per one standard deviation unit increase ^a in long-term pollutant concentrations

Pollutant		N	NE	N	EPI	N	DA
Ambient PM _{2.5}	Model 1 ^b	900	3.4 (-0.2, 7.2)	884	9.0 (3.2, 15)	896	4.5 (1.2, 7.9)
	Model 2 ^c	868	3.2 (-0.4, 7.0)	852	9.0 (3.2, 15.2)	864	4.4 (1.0, 7.9)
	Model 3 ^d	834	2.5 (-1.2, 6.3)	819	9.1 (3.2, 15.3)	830	4.4 (1.0, 7.9)
Individual PM _{2.5}	Model 1	860	2.4 (-1.1, 6.0)	847	7.4 (1.9, 13.2)	856	3.8 (0.5, 7.1)
	Model 2	830	2.1 (-1.5, 5.7)	817	6.6 (1.0, 12.5)	826	3.5 (0.2, 6.9)
	Model 3	800	1.2 (-2.5, 4.9)	788	6.3 (0.6, 12.2)	796	3.5 (0.1, 7.0)
NO _x	Model 1	900	4.6 (1.0, 8.4)	884	7.3 (1.7, 13.2)	896	1.4 (-1.8, 4.7)
	Model 2	868	3.8 (0.0, 7.7)	852	6.5 (0.6, 12.8)	864	1.1 (-2.3, 4.6)
	Model 3	834	3.5 (-0.4, 7.6)	819	6.3 (0.3, 12.6)	830	1.5 (-2.0, 5.2)
NO ₂	Model 1	900	1.8 (-2.1, 5.8)	884	5.9 (-0.1, 12.3)	896	0.1 (-3.4, 3.7)
	Model 2	868	0.9 (-3.1, 4.9)	852	4.7 (-1.4, 11.1)	864	-0.5 (-4.2, 3.1)
	Model 3	834	0.3 (-3.8, 4.5)	819	5.2 (-1.0, 11.7)	830	-0.3 (-4.0, 3.4)
Black Carbon	Model 1	903	1.8 (-1.4, 5.1)	887	2.8 (-2.1, 7.8)	899	-0.8 (-3.7, 2.1)
	Model 2	871	1.2 (-2.1, 4.6)	855	2.1 (-3.1, 7.3)	867	-1.3 (-4.4, 1.7)
	Model 3	837	0.9 (-2.5, 4.4)	822	1.1 (-4.1, 6.4)	833	-0.9 (-4.1, 2.2)

^a Standard deviation increases are as follows: 2.1 µg/m³ for ambient and individual PM_{2.5}, 17.5 ppb for NO_x, 4.8 ppb for NO₂ and 0.2 10⁻⁵/m for black carbon.

^b Model 1 is adjusted for age, race/ethnicity, gender, site.

^c Model 2 is adjusted for all the variables in model 1 plus SES variables: education, employment status, income-wealth index, neighborhood disadvantage index.

^d Model 3 is adjusted for all the variables in model 2 plus smoking, second-hand smoke exposure, BMI, alcohol consumption, emotional support, chronic burden and medication use. These estimates correspond to those shown in Figure 3.

Table S2: Percent differences in stress hormones per one standard deviation $\mu\text{g}/\text{m}^3$ increase ^a in short-term $\text{PM}_{2.5}$ concentrations ^b

Short-term $\text{PM}_{2.5}$ concentrations		N	NE	N	EPI	N	DA
Lag0	Model 1 ^c	921	0.2 (-2.9, 3.3)	905	0.5 (-4.2, 5.3)	917	2.1 (-0.7, 4.9)
	Model 2 ^d	889	0.3 (-2.8, 3.4)	873	0.7 (-4, 5.6)	885	1.7 (-1.2, 4.5)
	Model 3 ^e	854	-0.2 (-3.3, 3.0)	839	1.1 (-3.7, 6.0)	850	1.1 (-1.8, 4.1)
Lag1	Model 1	909	-1.9 (-5.1, 1.1)	893	-3.1 (-7.9, 1.5)	905	-1.4 (-4.3, 1.4)
	Model 2	878	-1.9 (-5.1, 1.2)	862	-2.9 (-7.9, 1.8)	874	-1.6 (-4.6, 1.2)
	Model 3	844	-2.1 (-5.4, 1.1)	829	-2.2 (-7.2, 2.5)	840	-2.2 (-5.2, 0.8)
Lag1-2	Model 1	898	-1.6 (-4.7, 1.5)	882	-3.7 (-8.6, 0.9)	894	-1.9 (-4.8, 0.9)
	Model 2	867	-1.5 (-4.7, 1.7)	851	-3.6 (-8.6, 1.2)	863	-2.0 (-5, 0.9)
	Model 3	834	-1.5 (-4.8, 1.8)	819	-2.5 (-7.5, 2.3)	830	-2.5 (-5.6, 0.4)
Lag1-3	Model 1	894	-2.0 (-5.2, 1.1)	878	-3.9 (-8.9, 0.8)	890	-1.7 (-4.6, 1.1)
	Model 2	863	-1.8 (-5.0, 1.4)	847	-3.7 (-8.7, 1.2)	859	-1.7 (-4.7, 1.2)
	Model 3	830	-1.9 (-5.2, 1.4)	815	-2.9 (-7.9, 2.0)	826	-2.2 (-5.2, 0.8)
Lag1-4	Model 1	891	-2.7 (-5.9, 0.4)	875	-5.0 (-10.0, -0.2)	887	-2.1 (-5, 0.7)
	Model 2	860	-2.4 (-5.7, 0.8)	844	-4.7 (-9.9, 0.2)	856	-2.3 (-5.3, 0.6)
	Model 3	827	-2.5 (-5.9, 0.7)	812	-3.8 (-8.9, 1.1)	823	-2.7 (-5.8, 0.3)
Lag1-5	Model 1	884	-2.2 (-5.4, 0.9)	868	-4.1 (-9.0, 0.7)	880	-1.6 (-4.5, 1.3)
	Model 2	854	-1.8 (-5.1, 1.4)	838	-3.8 (-8.9, 1.1)	850	-1.8 (-4.8, 1.1)
	Model 3	821	-2.1 (-5.4, 1.2)	806	-3.0 (-8.1, 1.9)	817	-2.1 (-5.1, 0.9)

^a Standard deviation increases are as follows: day of urine collection 8.3, day prior 7.9, 2 day average 7.1, 3-day average 6.5, 4-day average 6.0 and 5-day average 5.6 $\mu\text{g}/\text{m}^3$.

^b Short-term concentrations are pre-adjusted for calendar time splines (12 degrees of freedom (df)/year), temperature (6 df/year) and relative humidity (6 df/year) and an indicator variable for day of the week.

^c Model 1 is adjusted for age, race/ethnicity, gender, site.

^d Model 2 is adjusted for all the variables in model 1 plus education, employment status, income-wealth index and neighborhood disadvantage index.

^e Model 3 is adjusted for all the variables in model 2 plus smoking, second hand smoke exposure, BMI, alcohol consumption, emotional support, chronic burden, and medication use. These estimates correspond to those shown in Table 2.

Table S3: Percent difference in catecholamines per 1 standard deviation unit increase ^a in short-term pollutant concentrations adjusting for activities and feelings on the day of urine collection (acute stressors) ^b

Short-term PM2.5 concentration ^c	N	NE	N	EPI	N	DA
Lag0	720	-0.8 (-4.2, 2.5)	707	0.6 (-4.3, 5.8)	717	0.7 (-2.3, 3.8)
Lag1	709	-2.5 (-6.1, 0.9)	697	-2.2 (-7.5, 2.9)	706	-2.9 (-6.2, 0.3)
Lag1-2	702	-1.3 (-4.8, 2.1)	690	-2.0 (-7.3, 3.0)	699	-2.9 (-6.2, 0.3)
Lag1-3	699	-1.1 (-4.6, 2.3)	687	-1.8 (-7.1, 3.2)	696	-2.2 (-5.5, 0.9)
Lag1-4	696	-1.1 (-4.6, 2.4)	684	-2.4 (-7.9, 2.7)	693	-2.7 (-6.1, 0.5)
Lag1-5	692	-0.6 (-4.2, 2.9)	680	-1.9 (-7.4, 3.3)	689	-2.3 (-5.7, 0.9)

^a Standard deviation increases are as follows: day of urine collection 8.3, day prior 7.9, 2 day average 7.1, 3-day average 6.5, 4-day average 6.0 and 5-day average 5.6 µg/m³.

^b Models are adjusted for the following variables assessed on the day of urine collection: worked outside the home, exercised, smoked cigarettes, alcohol use, any medication use, felt happy, excited or content, felt worried, anxious or fearful and level of business, pressure or stress during the day. In addition models were also adjusted for age, race/ethnicity, gender, site, education, employment status, income-wealth index, neighborhood disadvantage index, second-hand smoke exposure, BMI, diuretic or sympathetic medication use, emotional support and chronic burden.

^c Short-term concentrations are adjusted for calendar time splines (12 degrees of freedom (df) /year), temperature (6 df/year) and relative humidity (6 df/year) and an indicator variable for day of the week.

Table S4: Percent difference in catecholamines per 1 standard deviation unit increase ^a in short-term pollutant concentrations when using 6 and 8 degrees of freedom for calendar time splines in the pre-adjustment model ^{b,c}

	NE		EPI		DA	
Short-term PM2.5 concentration	6 df	8 df	6 df	8 df	6 df	8 df
Lag0	-0.1 (-3.3, 3.0)	0.0 (-3.2, 3.0)	1.0 (-3.6, 5.9)	1.0 (-3.6, 5.9)	1.8 (-1.1, 4.7)	1.8 (-1.0, 4.8)
Lag1	-2.2 (-5.5, 1.0)	-2.1 (-5.4, 1.1)	-2.4 (-7.4, 2.3)	-2.0 (-7.0, 2.7)	-1.6 (-4.6, 1.3)	-1.5 (-4.5, 1.4)
Lag1-2	-1.6 (-4.8, 1.6)	-1.4 (-4.7, 1.8)	-2.8 (-7.8, 2)	-2.3 (-7.3, 2.4)	-2 (-5.0, 0.9)	-1.9 (-4.9, 1.1)
Lag1-3	-1.9 (-5.3, 1.3)	-1.8 (-5.2, 1.4)	-3.1 (-8.2, 1.7)	-2.6 (-7.7, 2.2)	-1.6 (-4.7, 1.3)	-1.6 (-4.6, 1.4)
Lag1-4	-2.6 (-5.9, 0.7)	-2.5 (-5.9, 0.7)	-4.0 (-9.2, 0.9)	-3.5 (-8.7, 1.3)	-2.2 (-5.2, 0.8)	-2.1 (-5.1, 0.9)
Lag1-5	-2.1 (-5.5, 1.1)	-2.1 (-5.5, 1.1)	-3.4 (-8.5, 1.4)	-2.9 (-8.0, 1.9)	-1.5 (-4.6, 1.4)	-1.5 (-4.6, 1.4)

^a Standard deviation increases are as follows: day of urine collection 8.3, day prior 7.9, 2 day average 7.1, 3-day average 6.5, 4-day average 6.0 and 5-day average 5.6 µg/m³.

^b Short-term concentrations are adjusted for calendar time splines (either 6 or 8 degrees of freedom (df) /year), temperature (6 df/year) and relative humidity (6 df/year) and an indicator variable for day of the week.

^c Models are adjusted for the following variables: age, race/ethnicity, gender, site, education, employment status, income-wealth index, neighborhood disadvantage index, second-hand smoke exposure, BMI, diuretic or sympathetic medication use, social support and chronic burden.

Table S5: Percent difference in catecholamines per 1 standard deviation unit increase ^a in short-term pollutant concentrations when adjusting for long-term PM_{2.5} concentrations ^b

Short-term PM2.5 concentration ^c	NE	EPI	DA
Lag0	-0.2 (-3.4, 3.0)	-0.1 (-5.0, 4.5)	1.2 (-1.6, 4.1)
Lag1	-1.9 (-5.2, 1.3)	-2.4 (-7.4, 2.3)	-1.9 (-4.9, 1.0)
Lag1-2	-1.3 (-4.6, 2.0)	-2.2 (-7.3, 2.5)	-2.3 (-5.3, 0.7)
Lag1-3	-1.5 (-4.9, 1.7)	-1.8 (-6.9, 3.0)	-1.8 (-4.9, 1.1)
Lag1-4	-2.0 (-5.4, 1.3)	-2.3 (-7.4, 2.5)	-2.3 (-5.4, 0.7)
Lag1-5	-1.7 (-5.1, 1.6)	-1.7 (-6.8, 3.2)	-1.8 (-4.9, 1.2)

^a Standard deviation increases are as follows: day of urine collection 8.3 µg/m³, day prior 7.9 µg/m³, 2 day average 7.1 µg/m³, 3-day average 6.5 µg/m³, 4-day average 6.0 µg/m³ and 5-day average 5.6 µg/m³.

^b Models are adjusted for the following variables: long-term (annual average) ambient PM_{2.5} concentration, age, race/ethnicity, gender, site, education, employment status, income-wealth index, neighborhood disadvantage index, second-hand smoke exposure, BMI, diuretic or sympathetic medication use, social support and chronic burden.

^c Short-term concentrations are pre-adjusted for calendar time splines (12 degrees of freedom (df) /year), temperature (6 df/year) and relative humidity (6 df/year) and an indicator variable for day of the week.

Table S6: Percent difference in catecholamines per 1 standard deviation unit increase ^a in long-term pollutant concentrations when additionally adjusting for season ^b

	NE	EPI	DA
Ambient PM _{2.5}	2.7 (-1.0, 6.6)	9.1 (3.2, 15.3)	4.6 (1.1, 8.2)
Individual PM _{2.5}	1.3 (-2.3, 5.0)	6.1 (0.5, 12.1)	3.6 (0.2, 7.1)
NO _x	3.5 (-0.4, 7.6)	6.2 (0.2, 12.6)	1.4 (-2.1, 5.0)
NO ₂	0.3 (-3.7, 4.5)	5.2 (-1.0, 11.8)	-0.3 (-4.0, 3.3)
Black Carbon	1.0 (-2.4, 4.5)	1.2 (-3.9, 6.5)	-0.8 (-3.9, 2.3)

^a Standard deviation increases are as follows: 2.1 µg/m³ for ambient and individual PM_{2.5}, 17.5 ppb for NO_x, 4.8 ppb for NO₂ and 0.2 10⁻⁵/m for black carbon.

^b Models were adjusted for the following variables: age, race/ethnicity, gender, site, education, employment status, income-wealth index, neighborhood disadvantage index, second-hand smoke exposure, BMI, diuretic or sympathetic medication use, social support, chronic burden and season of urine collection.

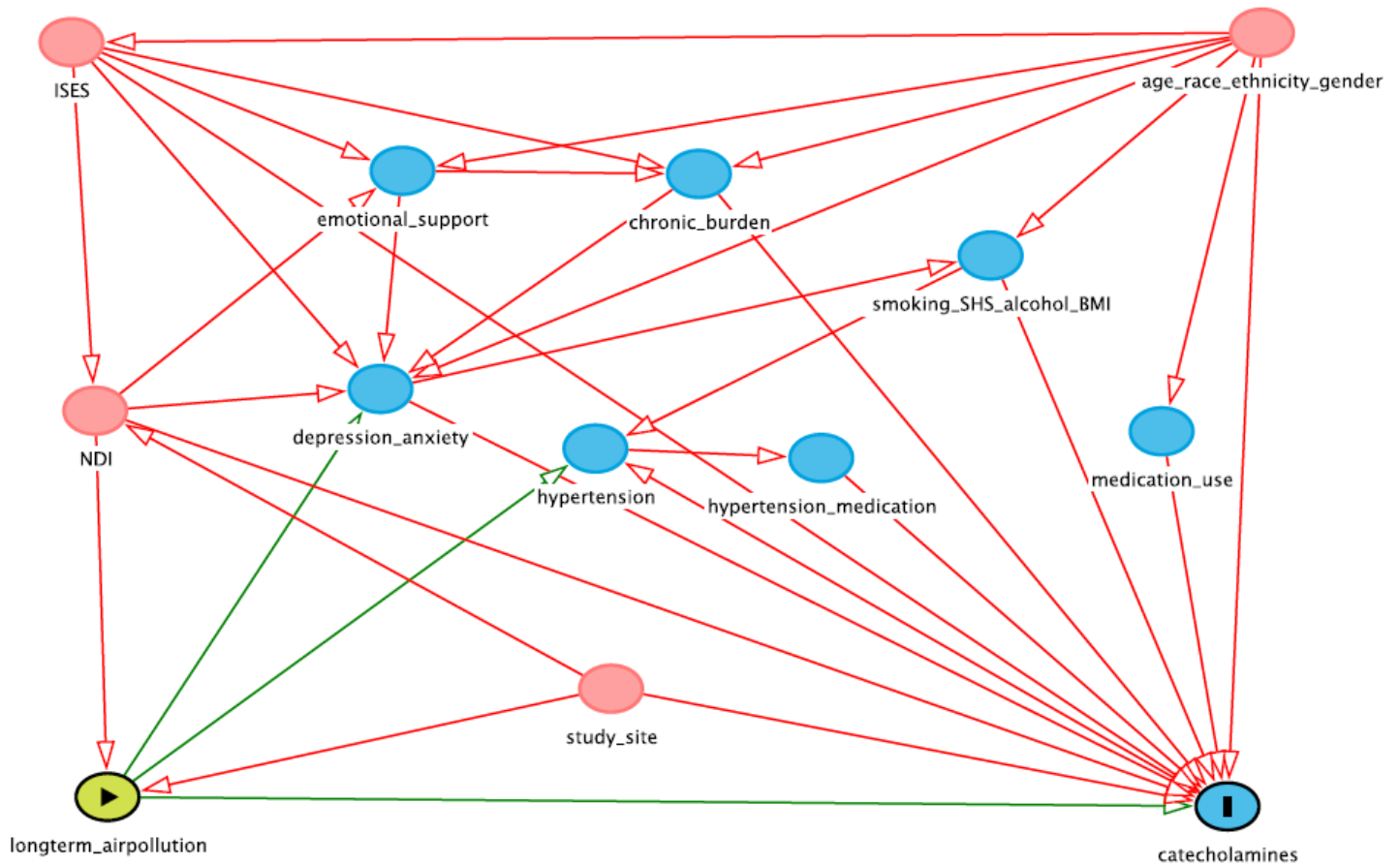


Figure S1: Directed Acyclic Graph (DAG) of association between air pollution and catecholamines. Abbreviations: ISES = individual socioeconomic status includes education, income-wealth index and employment status; NDI = neighborhood deprivation index; SHS = second-hand smoke; BMI = body mass index